# AI-Based Electricity Demand Intelligence — Explainable AI (XAI) System

**Component 4 — Explainable AI (XAI) Module** Final Year Research Project | Sri Lanka Institute of Information Technology (SLIIT)

## Research Information

| | |
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| **Research Component** | Component 4 — Explainable AI (XAI) for Electricity Demand Intelligence |
| **Academic Year** | 2025 / 2026 |

## Abstract

This research component develops an Explainable AI (XAI) system for electricity demand intelligence, providing interpretable justifications for an XGBoost-based demand forecasting model. The system applies two complementary explainability techniques — SHAP (SHapley Additive exPlanations) for global temporal feature importance analysis and LIME (Local Interpretable Model-agnostic Explanations) for local instance-level diagnosis — and presents both through an interactive React dashboard.

The XAI system enables energy analysts and decision-makers to understand not only what the AI model predicts, but why, by revealing which features (temperature, economic factors, temporal patterns) drive demand at a global scale over time and at the level of individual anomalous demand events.

## Research Problem & Motivation

Machine learning models for electricity demand forecasting often function as black boxes, producing accurate predictions without justifying their reasoning. This lack of transparency reduces trust and limits adoption in high-stakes energy management decisions. Stakeholders need to understand:

- Which factors most influence electricity demand — and how that influence shifts over time
- Why a specific day is classified as anomalous or showing unusual demand behaviour
- Whether the model's reasoning aligns with domain knowledge before trusting its predictions

This component addresses these gaps by integrating SHAP for temporal drift analysis across six years of data (2020–2025) and LIME for per-instance local explanations across 15 representative demand instances spanning low, medium, high, and peak demand levels.

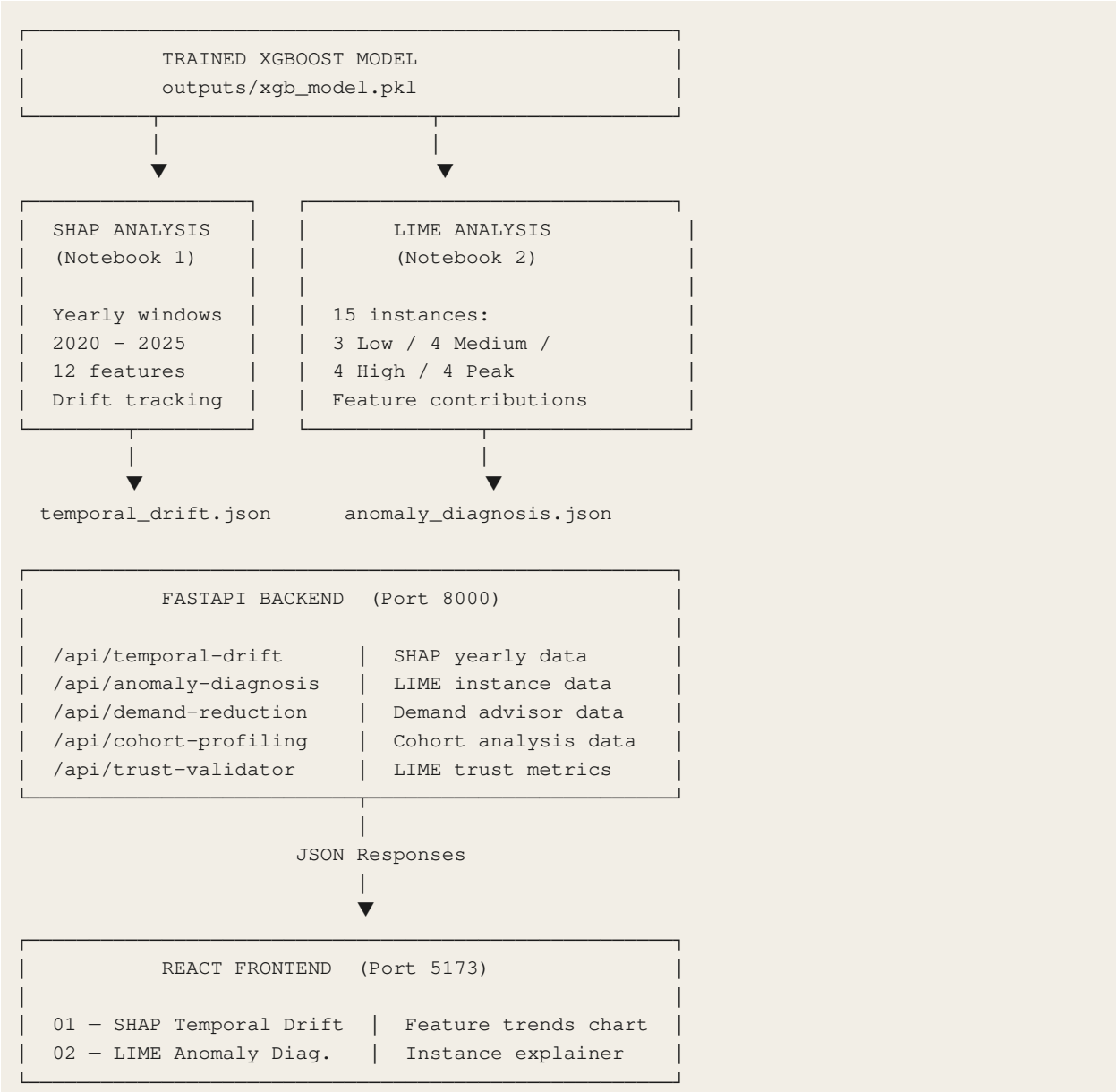
## Research Objectives

1. Apply SHAP to quantify global feature importance for the XGBoost demand forecasting model across yearly time windows (2020–2025)
2. Detect temporal drift by tracking how feature importances shift year-on-year

3. Apply LIME to generate local, instance-specific explanations for individual demand predictions
4. Categorise LIME explanations across four demand levels: Low, Medium, High, and Peak
5. Deploy both analyses through a REST API and an interactive two-panel dashboard
6. Enable integration with external components via URL parameters (e.g. from Behaviour Intelligence dashboard)

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## System Architecture




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## Technology Stack

### Backend

| Technology | Role |
|---|---|
| Python 3.9+ | Core language |
| FastAPI | REST API framework |
| Uvicorn | ASGI server |

| | |
|---|---|
| XGBoost | Primary forecasting model (pre-trained) |
| SHAP | Global feature importance & temporal drift analysis |
| LIME | Local instance-level explanation generation |
| scikit-learn | Preprocessing and supporting ML operations |
| pandas | Data processing |
| numpy | Numerical computation |
| joblib | Model loading |
| Optuna | Hyperparameter optimisation (model training) |

### Frontend

| Technology | Version | Role |
|---|---|---|
| React | 18.2.0 | UI component framework |
| Vite | 5.1.0 | Build tool and dev server |
| React Router DOM | 6.22.0 | Page navigation |
| Recharts | 2.12.0 | Interactive charts (line, bar) |
| Axios | 1.6.7 | HTTP client for API calls |
| Lucide React | 1.14.0 | Icon library |

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## XAI Methods

### SHAP — Global Temporal Drift Analysis

SHAP (SHapley Additive exPlanations) quantifies how much each input feature contributes to model output across the entire dataset. In this system, SHAP values are computed per year (2020–2025) to track how feature importance evolves over time.

| Analysis | Description |
|---|---|
| **Yearly feature importance** | Mean absolute SHAP values per feature per year |
| **Temporal drift** | Year-on-year change in feature rankings |
| **Feature trends** | Rising vs. declining feature influence over 6 years |
| **Yearly rankings** | Top 5 demand drivers identified per year |

### LIME — Local Anomaly Diagnosis

LIME (Local Interpretable Model-agnostic Explanations) fits a simple linear model around a specific prediction instance to explain why the model made a particular prediction for that individual input.

| Analysis | Description |
|---|---|
| **15 instances** | 3 Low, 4 Medium, 4 High, 4 Peak demand instances |
| **Feature contributions** | Signed weights showing positive/negative impact per feature |
| **Local fit $R^2$** | Confidence score of the local linear approximation |
| **Intercept** | Baseline demand before feature contributions |
| **Actual vs Predicted** | Side-by-side comparison with error |

## Project Structure

```
madushani_xai/
│
├── backend/
│   ├── app.py                          # FastAPI application (all 5 endpoints)
│   ├── requirements.txt                # Python dependencies
│   ├── generate_anomaly_diagnosis.py   # Script to generate LIME explanations
│   └── outputs/
│       ├── xgb_model.pkl               # Trained XGBoost model
│       ├── anomaly_diagnosis.json      # Pre-computed LIME explanations (15 instances)
│       ├── temporal_drift.json         # Pre-computed SHAP yearly drift data
│       └── demand_reduction.json       # SHAP demand reduction advisor data
│
├── frontend/
│   ├── src/
│   │   ├── App.jsx                     # Root component with sidebar navigation
│   │   ├── App.css                     # App-level styles
│   │   ├── main.jsx                    # Vite entry point
│   │   ├── index.css                   # Global styles
│   │   ├── components/                 # Shared UI components
│   │   └── pages/
│   │       ├── TemporalDrift.jsx       # SHAP temporal drift dashboard
│   │       ├── TemporalDrift.css       # Temporal drift styles
│   │       ├── LimeAnomalyDiagnosis.jsx # LIME anomaly diagnosis dashboard
│   │       ├── LimeAnomalyDiagnosis.css # LIME page styles
│   │       └── FeatureInteractions.css  # Feature interaction styles
│   ├── public/
│   ├── package.json
│   ├── vite.config.js                  # Vite config (proxy: /api → port 8000)
│   └── index.html
│
├── notebooks/
│   ├── 1_shap_temporal_drift.ipynb     # SHAP analysis across 2020–2025
│   └── 2_lime_anomaly_diagnosis.ipynb  # LIME explanations for 15 instances
│
└── README.md
```

## Setup & Installation

### Prerequisites

- Python 3.9 or higher
- Node.js 18 or higher
- npm 9 or higher

### 1. Clone the Repository

```
git clone <repository-url>
cd madushani_xai
```

### 2. Install Backend Dependencies

```
cd backend
pip install -r requirements.txt
```

### 3. Run SHAP Analysis (Notebook 1)

```
cd ../notebooks
jupyter notebook 1_shap_temporal_drift.ipynb
```

This will: - Load the trained XGBoost model - Compute SHAP values per year (2020–2025) for 12 features - Generate yearly rankings and drift statistics - Save results to `backend/outputs/temporal_drift.json`

### 4. Run LIME Anomaly Diagnosis (Notebook 2)

```
jupyter notebook 2_lime_anomaly_diagnosis.ipynb
```

This will: - Load the trained XGBoost model - Generate LIME explanations for 15 representative demand instances - Cover all demand levels: Low (3), Medium (4), High (4), Peak (4) - Save results to `backend/outputs/anomaly_diagnosis.json`

### 5. Start the Backend API

```
cd ../backend
uvicorn app:app --reload --host 0.0.0.0 --port 8000
```

### 6. Install and Start the Frontend

```
cd ../frontend
npm install
npm run dev
```

### 7. Access the System

| Service | URL |
|---|---|
| React Dashboard | http://localhost:5173 |
| API Base URL | http://127.0.0.1:8000 |
| Swagger API Docs | http://127.0.0.1:8000/docs |
| Health Check | http://127.0.0.1:8000 |

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## API Reference

Backend runs on port **8000**.

| Method | Endpoint | Description |
|---|---|---|
| `GET` | `/` | Health check / status |
| `GET` | `/api/temporal-drift` | SHAP feature importance per year (2020–2025) with rankings and drift |
| `GET` | `/api/anomaly-diagnosis` | 15 LIME explanations across all demand levels with feature contributions |
| `GET` | `/api/demand-reduction` | SHAP-based demand reduction advisor data |
| `GET` | `/api/cohort-profiling` | SHAP cohort-based analysis data |
| `GET` | `/api/trust-validator` | LIME trust and confidence metrics |

Interactive API documentation (Swagger UI): **http://127.0.0.1:8000/docs**

## Dashboard Features

### 01 — SHAP Temporal Drift

| Feature | Description |
|---|---|
| **KPI Cards** | Most Dominant Feature, Biggest Gainer, Biggest Decline, Years Analysed |
| **Feature Trend Line Chart** | SHAP importance over time per feature (2020–2025); toggleable per feature |
| **Yearly Ranking Cards** | Top 5 demand drivers identified for each year |
| **Drift Bar Chart** | Feature-level drift from 2020 to 2025 (blue = growth, red = decline) |
| **Interpretation Guide** | How to read SHAP values, rising/falling trends, and feature crossings |

### 02 — LIME Anomaly Diagnosis

| Feature | Description |
|---|---|
| **Stats Row** | Total instances (15), Mean Demand, Std Demand, Peak Threshold (p90) |
| **Instance List (Left Panel)** | Filterable by demand level (All / Low / Medium / High / Peak); shows timestamp, badge, $R^2$, actual vs predicted |
| **Explanation Panel (Right Panel)** | Feature contributions as signed horizontal bars (green = positive, red = negative); local fit $R^2$ confidence bar; intercept value |
| **Integration Hook** | Accepts `?source=bi&date=YYYY-MM-DD` URL params to auto-select instance from Behaviour Intelligence dashboard |

## Notebooks

| Notebook | Purpose |
|---|---|
| `1_shap_temporal_drift.ipynb` | Compute SHAP values per year, extract rankings, detect temporal drift, export JSON |
| `2_lime_anomaly_diagnosis.ipynb` | Generate LIME explanations for 15 instances across all demand levels, export JSON |

## Integration with Other Components

The LIME Anomaly Diagnosis page supports direct deep-linking from **Component 1 (Behaviour Intelligence)**. When a user clicks the XAI button for an abnormal day on the Behaviour Intelligence dashboard, it navigates to this component with the date pre-selected:

```
http://localhost:5173/lime-anomaly-diagnosis?source=bi&date=2023-07-15
```

The dashboard will automatically filter and highlight the matching instance for that date.

## License

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